

## Abstract / Project Goal

Our team was paired with a 53 year old veteran suffering from multiple health conditions including severe pedal edema. He is bound to a mobility scooter because it is painful and difficult to walk. Our solution is a therapy device that allows people with pedal edema to increase flexion-extension range of motion, increase blood and fluid flow, and strengthen their lower leg muscles. Unlike existing solutions, our device is automated, brings legs to a higher degree of extension, and does not require the leg to be fastened in place.

## Project Background

**Client:** 53 year old veteran suffering from different health conditions including pedal edema which caused him to be reliant to a mobility scooter.

**Pedal Edema:** type of edema that causes swelling in the lower legs or ankles. This condition occurs due to the accumulation of fluid inside the capillaries which increases the size and causes tissues to swell.



## Current Solutions

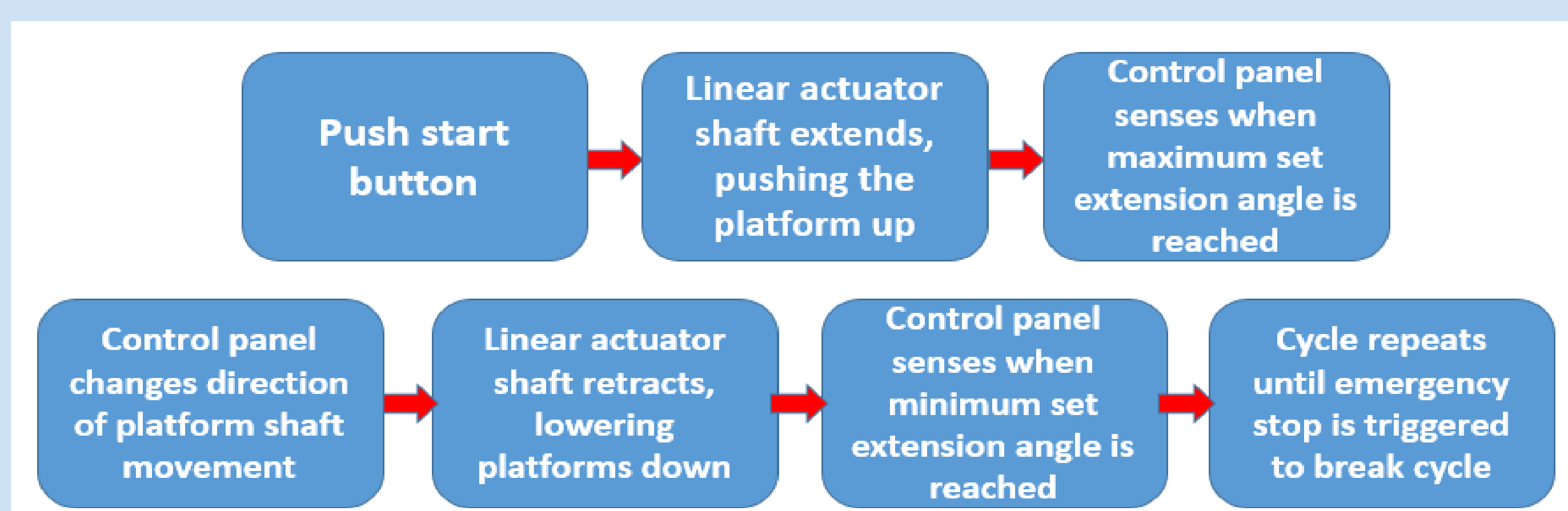


From left to right: Sequential compression wraps, elevation cushion, and Knee MD

**Limitations** of current devices include:

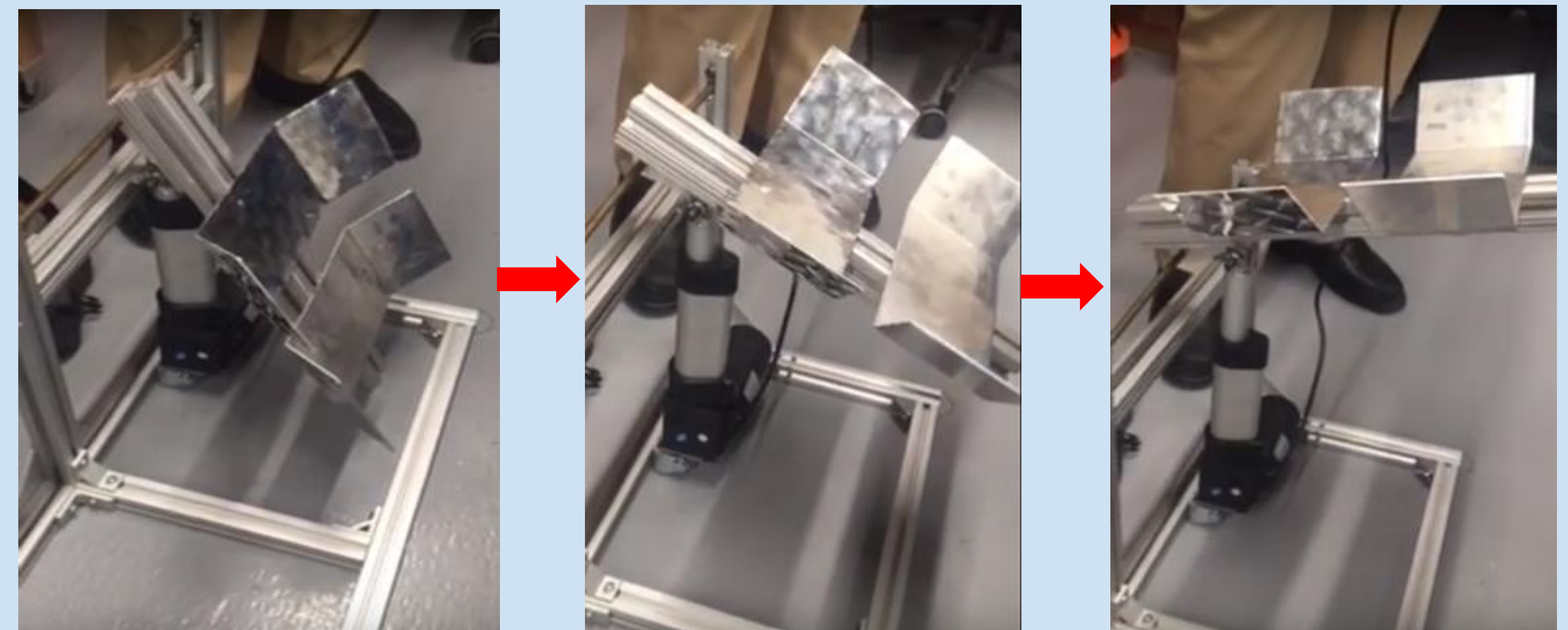
- High cost (upwards of \$3000)
- Require the ability to bend down or reach
- Incompatible with a mobile scooter
- Only target blood flow instead of blood flow and muscle strength

## Project Design Parameters



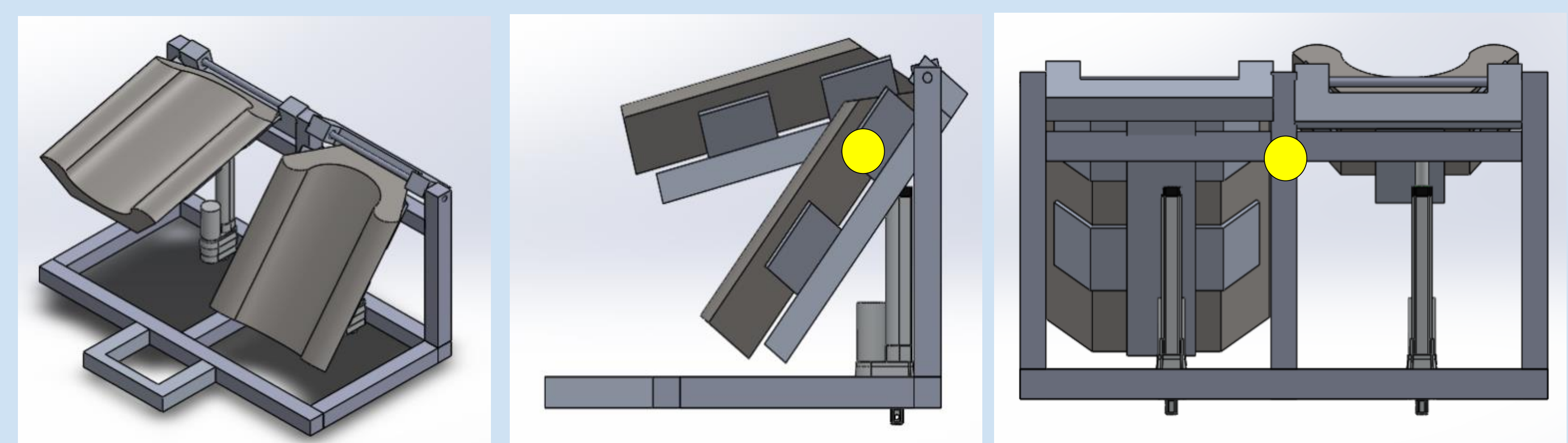
| Desired Feature                 | How It Is Achieved    |
|---------------------------------|-----------------------|
| Platform elevation/depression   | Linear actuator       |
| Easy/hard settings              | Load cells            |
| Device activation with mobility | Battery               |
| Support leg position            | Platforms             |
| Leg comfort                     | Foam Cushions         |
| Easy clean up                   | Faux leather material |
| Portability and safety          | Casters with locks    |

## Prototypes



Device motion (from left to right): **A)** platforms resting at and actuator fully retracted **B)** actuator extending vertically, elevating the platforms **C)** maximum actuator extension and platform elevation.

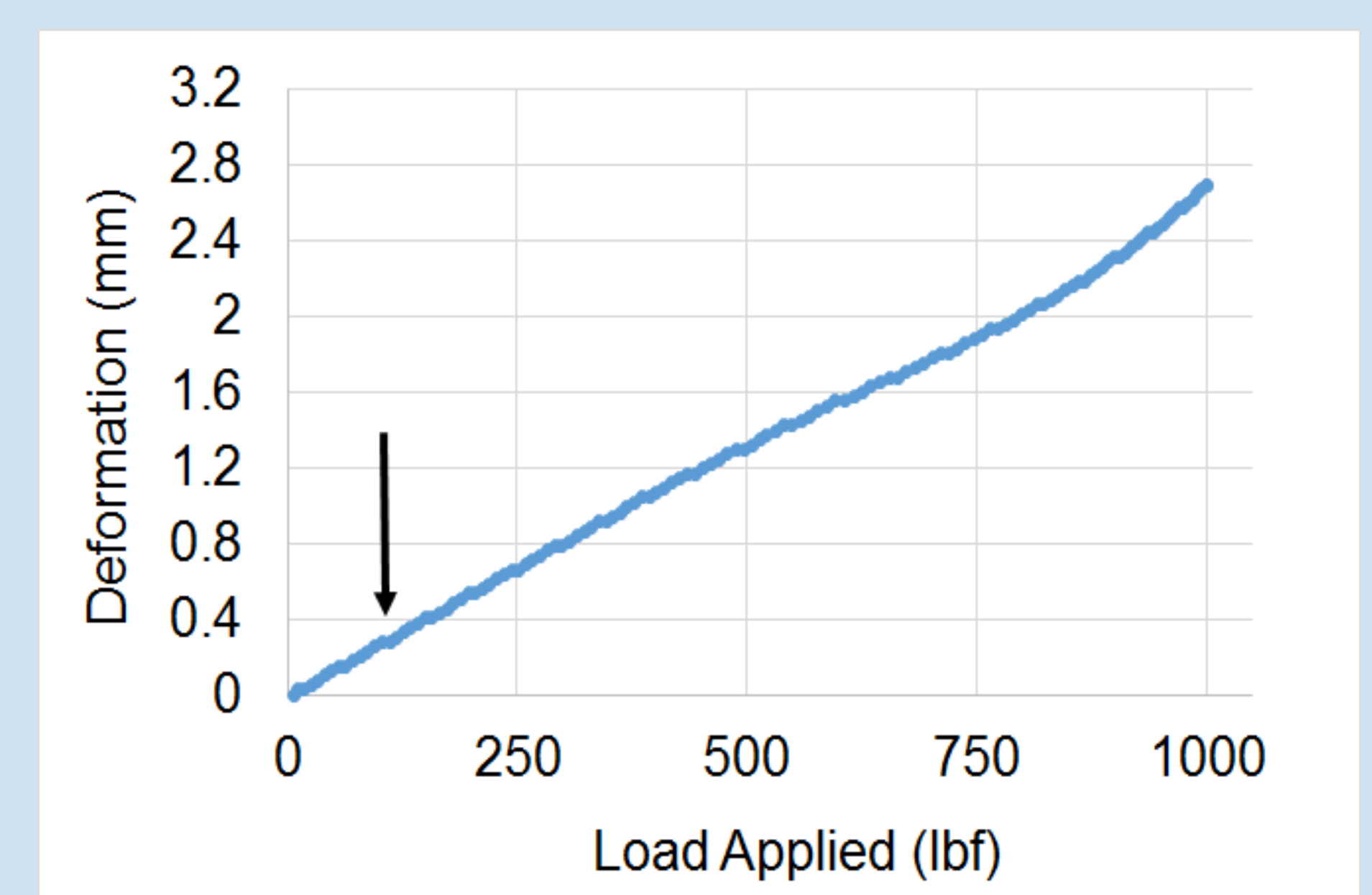
## Simulations



Device simulation (from left to right): overview, side view, and back view.

A center of mass simulation was performed using SolidWorks. The center of mass for our device (shown by yellow dot) is at x=9.91, y=24.57, and z=13.16 (inches).

## Testing and Results



Bending in platform is linear as load is applied. Allowable load (identified by arrow as 142 lbf) is far less than intended load upon use. Allowable load is the load required to produce 1/1000 of the extrusion length in bending deformation.

## Current and Future Work

**Current work:**

- Improve knee axial alignment
- Adjust speed settings
- Measure high pressure points
- Battery life test
- Ease of use test

**Future work:**

- Introduce resistance
- Device adjustability
- Add handles

## Acknowledgements

We would like to thank our collaborators in the mechanical engineering and nursing departments at the University of Detroit Mercy for their design contribution. We would also like to thank our client for his participation and inspiration for the device.